FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

4

IMPLEMENTING KNOWLEDGE REPRESENTATION, RULE-BASED REASONING (RBR), AND CASE-BASED REASONING (CBR)

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I. BACKGROUND INFORMATION

Knowledge Representation (KR) is the method of organizing facts, rules, and relationships so an AI system can understand and reason about the world.

Reasoning (R) allows the system to infer, make decisions, and solve problems based on that knowledge. Two major reasoning approaches from the module:

1. Rule-Based Reasoning (RBR)

Uses IF–THEN rules, a Knowledge Base, Working Memory, and an Inference Engine.

2. Case-Based Reasoning (CBR)

Solves new problems using past cases using the 4R Cycle:

Retrieve → Reuse → Revise → Retain

This laboratory lets you build small working implementations of both reasoning methods.

II. TECHNICAL ASSESSMENT

IMPLEMENTING KNOWLEDGE REPRESENTATION, RULE-BASED REASONING (RBR), AND CASE-BASED REASONING (CBR)

Objective

At the end of this laboratory activity, students are expected to:

1. Construct a simple knowledge base using Python data structures.
2. Implement a basic Rule-Based Reasoning (RBR) system using IF–THEN logic.
3. Build a simple Case-Based Reasoning (CBR) system using similarity matching.
4. Compare how RBR and CBR behave when solving the same type of problem.
5. Document observations and insights on KR, RBR, and CBR.

Materials Needed

1. Hardware
   - A laptop or desktop computer
   - Stable internet connection (for research, documentation, submission)
2. Software

A programming environment:

- Python (recommended: IDLE, VS Code, Jupyter Notebook, PyCharm)
- Any text editor (Notepad++, Sublime, VS Code) if needed

3. File Formats & Tools

Word processor for documentation:

- MS Word, Google Docs, or LibreOffice

- Screenshot tool (e.g., Snipping Tool, Lightshot, Print Screen)

General Instructions

1. This **Technical Assessment (TA3)** must be completed by Pair.
2. Include clear comments in your code to describe its logic.
3. Each program should display the correct input, process, and output.
4. Ensure that your script displays clear and readable outputs.
5. Screenshot of program output for RBR
6. Screenshot of program output for CBR
7. Reflection answers
8. Short description of your knowledge base and use case
9. Compile all Python codes and corresponding outputs into one PDF file.
10. Save it as **TA4_lastname1.pdf**
    (Example: TA2_Patilano.pdf)
11. Submit the **python file** and **PDF** file.

LABORATORY TASKS

PART 1 — Knowledge Representation (KR) Setup

Task 1A — Create a Knowledge Base

Create a Python dictionary storing at least:

- 5 facts about an object (e.g., weather, appliances, students, medical symptoms)
- 3 categories (e.g., temperature, status, presence, actions)

Example structure to follow:

```
facts = {
    "temperature": 32,
    "is_raining": False,
    "humidity": 85,
}
```

PART 2 — Implement Rule-Based Reasoning (RBR)

Task 2A — Create at least 3 IF–THEN rules

Write Python functions representing rules.

Example:

```
def rule_hot(facts):
    if facts["temperature"] > 30:
        return "Turn on the aircon"
```

Task 2B — Implement an Inference Engine

Create a loop that:

1. Reads the knowledge base
2. Checks each rule

3. Returns all actions inferred

Task 2C — Display Output

Example [Program should print:]

```
Rule-Based Reasoning Actions:
- Turn on the aircon
- Bring umbrella
```

PART 3 — Implement Case-Based Reasoning (CBR)

Task 3A — Build a Case Base

Create at least **3 past cases**, each containing:

- A set of problem features
- A matching solution

Example:

```
case_base = [
    {"problem": {"won't turn on": True, "noise": False},
     "solution": "Check power cable."}
]
```

Task 3C — Reuse Step

Return the solution of the most similar case.

Task 3D — Revise Step

Allow student to adjust solution manually or return unchanged.

Task 3E — Retain Step

Add the new case to the case base.

PART 4 — Comparative Reflection

Write a short paragraph answering:

1. Which system (RBR or CBR) was easier to implement? Why?

- RBR was easier to implement because it mainly required creating simple IF–THEN rules and checking whether the given facts satisfied the conditions. CBR was more challenging because it required creating several past cases, comparing the new problem with those cases, and selecting the most similar one before reusing its solution.

2. Which is more flexible in solving unstructured problems?

- CBR is more flexible in solving unstructured problems because it does not always require an exact predefined rule for every situation. Instead, it can look at previous cases with similar characteristics and use their solutions as a basis for handling a new problem, making it useful when problems can vary from one situation to another.

3. Which is better for situations with continuously changing data?

- CBR is better suited for situations with continuously changing data because new problems and solutions can be added to the case base as they occur. Over time, the system can build a larger collection of experiences and use more recent cases when solving future problems, while RBR would generally require its existing rules to be manually updated.

4. Examples of real-world applications for both.

- RBR can be applied to medical diagnosis, computer troubleshooting, and decision-support systems where specific conditions and actions can be represented using IF–THEN rules. CBR can be used in customer support, technical troubleshooting, and recommendation systems because these applications can compare a user's current problem with similar problems that were previously solved.

Your Code - Laboratory Tasks ⇩

Code:

```
# CS0065 - Intelligent Systems
# Technical Assessment 4
# Knowledge Representation, RBR, and CBR
# Use Case: Computer Troubleshooting


# ============================================================
# PART 1 - KNOWLEDGE REPRESENTATION
# ============================================================

print("=" * 50)
print("COMPUTER TROUBLESHOOTING SYSTEM")
print("=" * 50)

print("\nINPUT")
print("Enter the following information about the computer.")

computer_on = input("Is the computer turned on? (yes/no): ").lower()
power_connected = input("Is the power cable connected? (yes/no): ").lower()
screen_black = input("Is the screen black? (yes/no): ").lower()
battery_low = input("Is the battery low? (yes/no): ").lower()
internet_connected = input("Is the internet connected? (yes/no): ").lower()

# Knowledge Base with 3 categories
facts = {
    "system_status": {
```

```
        "computer_on": computer_on == "yes",
        "internet_connected": internet_connected == "yes"
    },
    "hardware": {
        "power_connected": power_connected == "yes",
        "battery_low": battery_low == "yes"
    },
    "symptoms": {
        "screen_black": screen_black == "yes"
    }
}

print("\nKnowledge Base created successfully.")


# ============================================================
# PART 2 - RULE-BASED REASONING
# ============================================================

print("\n" + "=" * 50)
print("RULE-BASED REASONING (RBR)")
print("=" * 50)

actions = []

# Rule 1
if not facts["system_status"]["computer_on"] and facts["hardware"]["power_connected"]:
    actions.append("Check the battery level or power source.")

# Rule 2
if facts["symptoms"]["screen_black"] and not facts["system_status"]["computer_on"]:
    actions.append("Check the monitor/display connection and power.")

# Rule 3
if facts["hardware"]["battery_low"]:
    actions.append("Charge the computer before troubleshooting further.")

print("\nPROCESS")
print("The inference engine checks the facts against the IF-THEN rules.")

print("\nOUTPUT")
if actions:
    for action in actions:
```

```
        print("- " + action)
else:
    print("- No rule was triggered.")


# ============================================================
# PART 3 - CASE-BASED REASONING
# ============================================================

print("\n" + "=" * 50)
print("CASE-BASED REASONING (CBR)")
print("=" * 50)

# Past cases
case_base = [
    {
        "problem": [False, True, True, True, False],
        "solution": "Check the battery level and power connection."
    },
    {
        "problem": [True, True, True, False, False],
        "solution": "Check the monitor cable and display settings."
    },
    {
        "problem": [True, True, False, False, False],
        "solution": "Restart the computer and check for software issues."
    },
    {
        "problem": [False, False, True, True, False],
        "solution": "Connect the power cable and charge the computer."
    },
    {
        "problem": [True, False, False, False, True],
        "solution": "Check the power adapter and network settings."
    }
]

# New problem
new_problem = [
    facts["system_status"]["computer_on"],
    facts["hardware"]["power_connected"],
    facts["symptoms"]["screen_black"],
    facts["hardware"]["battery_low"]
```

```
]

# RETRIEVE
best_case = None
best_score = -1

for case in case_base:
    score = sum(a == b for a, b in zip(new_problem, case["problem"]))

    if score > best_score:
        best_score = score
        best_case = case

print("\nRETRIEVE")
print("Most similar previous case found.")

# REUSE
solution = best_case["solution"]

print("\nREUSE")
print("Reusing solution:")
print("- " + solution)

# REVISE
print("\nREVISE")
print("Solution reviewed and kept unchanged.")

# RETAIN
case_base.append({
    "problem": new_problem,
    "solution": solution
})

print("\nRETAIN")
print("New case has been added to the case base.")

print("\n" + "=" * 50)
print("PROGRAM COMPLETED")
print("=" * 50)
```

Output

```
INPUT
Enter the following information about the computer.
Is the computer turned on? (yes/no): No
Is the power cable connected? (yes/no): Yes
Is the screen black? (yes/no): Yes
Is the battery low? (yes/no): Yes
Is the internet connected? (yes/no): No

Knowledge Base created successfully.

==================================================
RULE-BASED REASONING (RBR)
==================================================

PROCESS
The inference engine checks the facts against the IF-THEN rules.

OUTPUT
- Check the battery level or power source.
- Check the monitor/display connection and power.
- Charge the computer before troubleshooting further.

==================================================
CASE-BASED REASONING (CBR)
==================================================

RETRIEVE
Most similar previous case found.

REUSE
Reusing solution:
- Check the battery level and power connection.

REVISE
Solution reviewed and kept unchanged.
RETAIN
New case has been added to the case base.

==================================================
PROGRAM COMPLETED
```

```
INPUT
Enter the following information about the computer.
Is the computer turned on? (yes/no): yes
Is the power cable connected? (yes/no): yes
Is the screen black? (yes/no): yes
Is the battery low? (yes/no): no
Is the internet connected? (yes/no): no

Knowledge Base created successfully.

================================================
RULE-BASED REASONING (RBR)
================================================

PROCESS
The inference engine checks the facts against the IF-THEN rules.

OUTPUT
- No rule was triggered.

================================================
CASE-BASED REASONING (CBR)
================================================

RETRIEVE
Most similar previous case found.

REUSE
Reusing solution:
- Check the monitor cable and display settings.

REVISE
Solution reviewed and kept unchanged.

RETAIN
New case has been added to the case base.

================================================
PROGRAM COMPLETED
------------------------------------------------
```

```
INPUT
Enter the following information about the computer.
Is the computer turned on? (yes/no): Yes
Is the power cable connected? (yes/no): yes
Is the screen black? (yes/no): no
Is the battery low? (yes/no): no
Is the internet connected? (yes/no): no

Knowledge Base created successfully.

==================================================
RULE-BASED REASONING (RBR)
==================================================

PROCESS
The inference engine checks the facts against the IF-THEN rules.

OUTPUT
- No rule was triggered.

==================================================
CASE-BASED REASONING (CBR)
==================================================

RETRIEVE
Most similar previous case found.

REUSE
Reusing solution:
- Restart the computer and check for software issues.

REVISE
Solution reviewed and kept unchanged.

RETAIN
New case has been added to the case base.

==================================================
PROGRAM COMPLETED
--------------------------------------------------
```

```
Enter the following information about the computer.
Is the computer turned on? (yes/no): No
Is the power cable connected? (yes/no): No
Is the screen black? (yes/no): Yes
Is the battery low? (yes/no): Yes
Is the internet connected? (yes/no): No

Knowledge Base created successfully.

==================================================
RULE-BASED REASONING (RBR)
==================================================

PROCESS
The inference engine checks the facts against the IF-THEN rules.

OUTPUT
- Check the monitor/display connection and power.
- Charge the computer before troubleshooting further.

==================================================
CASE-BASED REASONING (CBR)
==================================================

RETRIEVE
Most similar previous case found.

REUSE
Reusing solution:
- Connect the power cable and charge the computer.

REVISE
Solution reviewed and kept unchanged.

RETAIN
New case has been added to the case base.

==================================================
PROGRAM COMPLETED
--------------------------------------------------
```

```
INPUT
Enter the following information about the computer.
Is the computer turned on? (yes/no): Yes
Is the power cable connected? (yes/no): No
Is the screen black? (yes/no): No
Is the battery low? (yes/no): No
Is the internet connected? (yes/no): Yes

Knowledge Base created successfully.

================================================
RULE-BASED REASONING (RBR)
================================================

PROCESS
The inference engine checks the facts against the IF-THEN rules.

OUTPUT
- No rule was triggered.

================================================
CASE-BASED REASONING (CBR)
================================================

RETRIEVE
Most similar previous case found.

REUSE
Reusing solution:
- Check the power adapter and network settings.

REVISE
Solution reviewed and kept unchanged.

RETAIN
New case has been added to the case base.

================================================
PROGRAM COMPLETED
------------------------------------------------
```